APPLY Hydroxide precipitates AQ-37

**Equation: $[Cu(H_2O)_6]^{2+}$ with NaOH(aq). [1]**



APPLY Hydroxide precipitates AQ-38

**Equation: $[Fe(H_2O)_6]^{2+}$ with NaOH(aq). [1]**



APPLY Hydroxide precipitates AQ-39

**Equation: $[Fe(H_2O)_6]^{3+}$ with NaOH(aq). [1]**



APPLY Hydroxide precipitates AQ-40

**Equation: $[Al(H_2O)_6]^{3+}$ with NaOH(aq). [1]**



APPLY Hydroxide precipitates AQ-41

**Equation: $[Cu(H_2O)_6]^{2+}$ with a little $NH_3$(aq). [1]**



APPLY Hydroxide precipitates AQ-42

**Equation: $[Fe(H_2O)_6]^{2+}$ with $NH_3$(aq). [1]**



APPLY Hydroxide precipitates AQ-43

**Equation: $[Fe(H_2O)_6]^{3+}$ with $NH_3$(aq). [1]**



APPLY Hydroxide precipitates AQ-44

**Equation: $[Al(H_2O)_6]^{3+}$ with $NH_3$(aq). [1]**

APPLY Hydroxide precipitates AQ-38

1. $[Fe(H_2O)_6]^{2+} + 2OH^- \rightarrow [Fe(H_2O)_4(OH)_2] + 2H_2O$



APPLY Hydroxide precipitates AQ-37

1. $[Cu(H_2O)_6]^{2+} + 2OH^- \rightarrow [Cu(H_2O)_4(OH)_2] + 2H_2O$

State symbols only if asked. Balanced is essential.



APPLY Hydroxide precipitates AQ-40

1. $[Al(H_2O)_6]^{3+} + 3OH^- \rightarrow [Al(H_2O)_3(OH)_3] + 3H_2O$



APPLY Hydroxide precipitates AQ-39

1. $[Fe(H_2O)_6]^{3+} + 3OH^- \rightarrow [Fe(H_2O)_3(OH)_3] + 3H_2O$



APPLY Hydroxide precipitates AQ-42

1. $[Fe(H_2O)_6]^{2+} + 2NH_3 \rightarrow [Fe(H_2O)_4(OH)_2] + 2NH_4^+$



APPLY Hydroxide precipitates AQ-41

1. $[Cu(H_2O)_6]^{2+} + 2NH_3 \rightarrow [Cu(H_2O)_4(OH)_2] + 2NH_4^+$

$NH_3$ takes $H^+$, so the by-product is $NH_4^+$, not $H_2O$.



APPLY Hydroxide precipitates AQ-44

1. $[Al(H_2O)_6]^{3+} + 3NH_3 \rightarrow [Al(H_2O)_3(OH)_3] + 3NH_4^+$



APPLY Hydroxide precipitates AQ-43

1. $[Fe(H_2O)_6]^{3+} + 3NH_3 \rightarrow [Fe(H_2O)_3(OH)_3] + 3NH_4^+$

APPLY Hydroxide precipitates AQ-45

**Explain why adding a base to a metal-aqua ion forms a precipitate. [2]**



APPLY Hydroxide precipitates not in notes AQ-46

**Write an equation for the green Fe(II) precipitate turning brown in air. [2]**



APPLY Excess ammonia AQ-47

**Equation: excess $NH_3$ with $[Cu(H_2O)_6]^{2+}$. [1]**



APPLY Excess ammonia AQ-48

**Equation: excess $NH_3$ with the copper hydroxide precipitate. [1]**



APPLY Excess ammonia not in notes AQ-49

**Explain why the co-ordination number stays 6 when $NH_3$ replaces $H_2O$ ligands. [2]**



APPLY Excess ammonia AQ-50

**$NH_3$ can act as a base or as a ligand. Explain both. [2]**



APPLY Amphoteric aluminium AQ-51

**Equation: aluminium hydroxide acting as a base (with acid). [1]**



APPLY Amphoteric aluminium AQ-52

**Equation: aluminium hydroxide acting as an acid (with excess $OH^-$). [1]**

APPLY Hydroxide precipitates not in notes AQ-46

1. $4[Fe(H_2O)_4(OH)_2] + O_2 \rightarrow 4[Fe(H_2O)_3(OH)_3] + 2H_2O$
2. Fe(II) is **oxidised** to Fe(III)

Not in your notes as an equation; the notes only state the change.



APPLY Hydroxide precipitates AQ-45

1. The base **removes $H^+$** from water ligands (deprotonation), turning them into $OH^-$ ligands
2. Until the complex is **neutral** (no charge), so it is **insoluble**



APPLY Excess ammonia AQ-48

1. $[Cu(H_2O)_4(OH)_2] + 4NH_3 \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 2OH^- + 2H_2O$

Check the charge: 0 on the left, 2+ and 2(−1) = 0 on the right.



APPLY Excess ammonia AQ-47

1. $[Cu(H_2O)_6]^{2+} + 4NH_3 \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 4H_2O$



APPLY Excess ammonia AQ-50

1. Base: accepts an **$H^+$** from a water ligand (forms $NH_4^+$)
2. Ligand: donates its **lone pair on N** to the metal ion (co-ordinate bond)



APPLY Excess ammonia not in notes AQ-49

1. $NH_3$ and $H_2O$ are **similar in size**
2. Both are **uncharged** (neutral) ligands

Your notes cover the substitution; this explanation is from the AQA spec (3.2.5).



APPLY Amphoteric aluminium AQ-52

1. $[Al(H_2O)_3(OH)_3] + OH^- \rightarrow [Al(OH)_4]^- + 3H_2O$

Also accepted: $\rightarrow [Al(H_2O)_2(OH)_4]^- + H_2O$ (only 1 water then).



APPLY Amphoteric aluminium AQ-51

1. $[Al(H_2O)_3(OH)_3] + 3H^+ \rightarrow [Al(H_2O)_6]^{3+}$

APPLY Amphoteric aluminium AQ-53

**Equation: $[Al(H_2O)_6]^{3+}$ with excess NaOH "all at once". [1]**



APPLY Amphoteric aluminium AQ-54

**Explain why aluminium hydroxide dissolves in excess NaOH but not in excess $NH_3$. [2]**



APPLY Carbonate AQ-55

**Equation: $[Cu(H_2O)_6]^{2+}$ with $Na_2CO_3(aq)$. [1]**



APPLY Carbonate AQ-56

**Equation: $[Fe(H_2O)_6]^{2+}$ with $Na_2CO_3(aq)$. [1]**



APPLY Carbonate AQ-57

**Equation: $[Fe(H_2O)_6]^{3+}$ with $Na_2CO_3(aq)$. [1]**



APPLY Carbonate AQ-58

**Equation: $[Al(H_2O)_6]^{3+}$ with $Na_2CO_3(aq)$. [1]**



APPLY Chloride ions AQ-59

**Equation: $[Cu(H_2O)_6]^{2+}$ with concentrated HCl. [1]**



APPLY Chloride ions AQ-60

**Explain why the co-ordination number changes from 6 to 4 when $Cl^-$ replaces $H_2O$. [2]**

APPLY Amphoteric aluminium AQ-54

1. NaOH provides a **high concentration of $OH^-$**, which removes a further $H^+$ to form soluble $[Al(OH)_4]^-$
2. $NH_3$ is a **weak base**: the $[OH^-]$ is too low to do this



APPLY Amphoteric aluminium AQ-53

1. $[Al(H_2O)_6]^{3+} + 4OH^- \rightarrow [Al(OH)_4]^- + 6H_2O$



APPLY Carbonate AQ-56

1. $[Fe(H_2O)_6]^{2+} + CO_3^{2-} \rightarrow FeCO_3 + 6H_2O$



APPLY Carbonate AQ-55

1. $[Cu(H_2O)_6]^{2+} + CO_3^{2-} \rightarrow CuCO_3 + 6H_2O$



APPLY Carbonate AQ-58

1. $2[Al(H_2O)_6]^{3+} + 3CO_3^{2-} \rightarrow 2[Al(H_2O)_3(OH)_3] + 3CO_2 + 3H_2O$



APPLY Carbonate AQ-57

1. $2[Fe(H_2O)_6]^{3+} + 3CO_3^{2-} \rightarrow 2[Fe(H_2O)_3(OH)_3] + 3CO_2 + 3H_2O$

Watch the charge on Fe: 3+. 2 × 3+ balances 3 × 2−.



APPLY Chloride ions AQ-60

1. $Cl^-$ ions are **larger** than $H_2O$ molecules
2. So **fewer fit** around the metal ion (only 4)



APPLY Chloride ions AQ-59

1. $[Cu(H_2O)_6]^{2+} + 4Cl^- \rightleftharpoons [CuCl_4]^{2-} + 6H_2O$

APPLY Chloride ions not in notes AQ-61

**How could you reverse the formation of $[CuCl_4]^{2-}$? [1]**



APPLY Acidity AQ-62

**Equation: $[Fe(H_2O)_6]^{3+}$ acting as an acid in water. [1]**



APPLY Acidity AQ-63

**Equation: $[Al(H_2O)_6]^{3+}$ acting as an acid in water. [1]**



APPLY Acidity AQ-64

**Equation: $[Fe(H_2O)_6]^{2+}$ acting as an acid in water. [1]**



APPLY Acidity AQ-65

**Explain why a solution of iron(III) chloride is acidic. [2]**



APPLY Colours AQ-66

**Why does $[Fe(H_2O)_6]^{3+}(aq)$ usually look yellow-brown, not purple? [1]**



APPLY Complex ions AQ-67

**Give the charge of each: $[Fe(H_2O)_5(OH)]^{x}$ and $[Al(OH)_4]^{y}$. [2]**



APPLY Identify & practical AQ-68

**A solution gives a white ppt with NaOH(aq) that dissolves in excess. Identify the ion and the final complex. [2]**

APPLY Acidity AQ-62

1. $[Fe(H_2O)_6]^{3+} + H_2O \rightleftharpoons [Fe(H_2O)_5(OH)]^{2+} + H_3O^+$

Also accepted: $[Fe(H_2O)_6]^{3+} \rightleftharpoons [Fe(H_2O)_5(OH)]^{2+} + H^+$



APPLY Chloride ions not in notes AQ-61

1. **Add water** (dilute it): the equilibrium shifts left, back to blue $[Cu(H_2O)_6]^{2+}$

Not in your notes, but follows from the ⇌ in the equation.



APPLY Acidity AQ-64

1. $[Fe(H_2O)_6]^{2+} + H_2O \rightleftharpoons [Fe(H_2O)_5(OH)]^{+} + H_3O^+$

Happens to a much smaller extent than for 3+ ions.



APPLY Acidity AQ-63

1. $[Al(H_2O)_6]^{3+} + H_2O \rightleftharpoons [Al(H_2O)_5(OH)]^{2+} + H_3O^+$



APPLY Colours AQ-66

1. Some is hydrolysed to **$[Fe(H_2O)_5(OH)]^{2+}$**, which is yellow-brown



APPLY Acidity AQ-65

1. Water ligands lose $H^+$ (the aqua ion is hydrolysed)
2. $[Fe(H_2O)_6]^{3+} + H_2O \rightleftharpoons [Fe(H_2O)_5(OH)]^{2+} + H_3O^+$, so **$H_3O^+$** ions are produced



APPLY Identify & practical AQ-68

1. **$Al^{3+}$**
2. **$[Al(OH)_4]^-$** (colourless solution)



APPLY Complex ions AQ-67

1. x = **2+** (3+ and one $OH^-$)
2. y = **1−** (3+ and four $OH^-$)

Charge = metal charge + (−1 for each $OH^-$ or $Cl^-$; 0 for $H_2O$ or $NH_3$).

APPLY Identify & practical AQ-69

**A blue solution gives a blue ppt with a little $NH_3$ and a deep blue solution with excess. Identify the ion and final complex. [2]**

APPLY Identify & practical AQ-69

1. **$Cu^{2+}$** ($[Cu(H_2O)_6]^{2+}$)
2. **$[Cu(NH_3)_4(H_2O)_2]^{2+}$**